

CHEMISTRY MODEL THEORY EXAM

Group: I and II

Time: 3 Hrs

Max. Marks: 75

PART A - (10 x 2 = 20 marks)

Answer ALL questions..

1. The carbonate hardness is called as temporary hardness. Why?
2. Differentiate between scale and sludge.
3. Name the catalyst used in cationic and anionic polymerization.
4. Distinguish between thermo and thermosetting plastics.
5. What is meant by emf series? Write its applications.
6. Calculate the reduction potential of a Pb electrode in solution of 0.018 M Pb^{2+} ions. [$E^0 = -0.13V$]
7. What is Metal cladding?
8. Write note on Oxidation corrosion.
9. How many components are present in the following system
(a) Water $\rightleftharpoons$ water vapour (b) KCl + Water $\rightleftharpoons$ KCl hydrate
10. What is condensed or reduced phase rule?

PART B -(5 x 11 = 55 marks)

UNIT I

11. (a) Describe demineralization process of water softening using Zeolite process. (6)
(b) Give a brief account on internal treatment process. (5)

Or

12. (a) How the water is desalinated by Reverse Osmosis process? (6)
(b) 20 ml of a hard water sample required 25.7 ml of 0.01 M EDTA solution. 20 ml of water sample after boiling and filtering required 14.2 ml of 0.01 M EDTA solution. Calculate its carbonate, non-carbonate and total hardness. (5)

UNIT II

13. (a) Explain the preparation, properties and uses of PVC and Silicone rubber. (6)
(b) Write a note on Ziegler Natta polymerization mechanism. (5)

Or

14. (a) Discuss briefly the types of polymerization (6)
(b) With a neat diagram, explain the Compression and Extrusion moulding of plastics. (5)

UNIT III

15. (a) What is EMF? How is it determined experimentally by potentiometric method. (6)
(b) Explain the Glass electrode (5)

Or

16. (a) What are fuel cells? How $H_2 - O_2$ fuel cell is constructed? (6)
(b) Explain construction, application & limitations of Hydrogen electrode. (5)

UNIT IV

17. (a) Explain the effects of the following factors on the rate of corrosion: (i) Nature of corrosion product (ii) pH (iii) Anodic and cathodic areas (iv) Hydrogen over voltage. (6)
(b) What are the main objectives of electroplating? Give an account of the method used in electroplating of nickel on steel. (5)

Or

18. (a) Explain galvanizing and tinning. In what way galvanizing is differed from tinning. (6)
(b) Discuss Pitting corrosion and Concentration cell corrosion (5)

UNIT V

19. (a) Draw and explain the phase diagram for water system. (6)
(b) Explain the Cu- Ni phase diagram. (5)

Or

20. (a) With a neat phase diagram explain the Pb-Ag system. (6)
(b) Define and Derive the phase rule equation. (5)